Lab 01
Software Engineering
Harshith J
PES1UG24CS185

Functional Requirements:

FR 001)
The system shall allow club members to set and update personal reading page goals.

FR 002)
The system shall allow club members to record and track their reading progress toward their page goals.

FR 003)
The system shall allow club members to create and participate in chapter based discussion threads.

FR 004)
The system shall allow Club Members to mark discussion comments as spoilers and hide spoiler content until the relevant chapter is read or the user chooses to reveal it.

FR 005)
The system shall allow Club Members to vote in monthly polls for selecting the next book, while preventing duplicate votes.

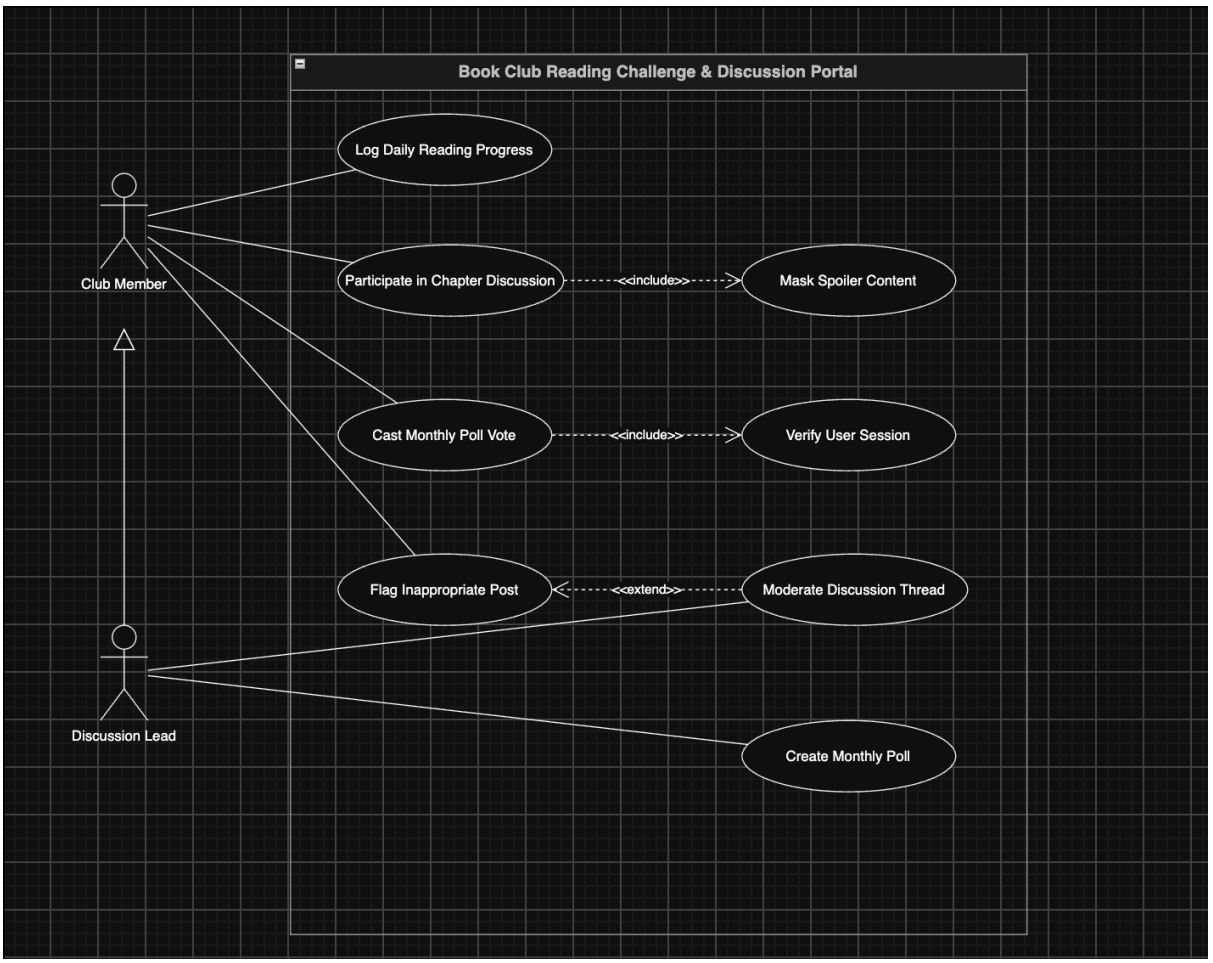
Complete Requirements Table:

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Club Members to set and update personal reading page goals.	High	Pass: Member can enter and modify a target number of pages. Fail: The system does not save the updated goal.	Allows members to establish and manage their personal reading targets.

FR-002	Functional	The system shall allow Club Members to record and track their reading progress toward their page goals.	High	Pass: Member can update pages read and view their progress toward the goal. Fail: Progress is not updated or displayed correctly.	Helps members monitor whether they are meeting their reading goals.
FR-003	Functional	The system shall allow Club Members to create and participate in chapter-based discussion threads.	High	Pass: Member can create a discussion comment for a chapter and view existing comments. Fail: Comments cannot be posted or viewed.	Facilitates community discussion around the book.
FR-004	Functional	The system shall allow Club Members to mark discussion comments as spoilers and hide spoiler content until the relevant chapter is read or the user chooses to reveal it.	High	Pass: Spoiler content is hidden and can only be revealed when permitted. Fail: Spoiler content is immediately visible.	Prevents members from accidentally seeing information from chapters they have not read.
FR-005	Functional	The system shall allow Club Members to vote in monthly polls for selecting the next book while preventing duplicate votes.	High	Pass: A member can cast one vote and cannot vote again in the same poll. Fail: The member can submit multiple votes.	Ensures fair monthly book selection.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The system shall update monthly poll voting results in real time while maintaining secure user-session verification.	High	Pass: Vote results update within the defined target latency and only authenticated sessions can submit votes. Fail: Results are significantly delayed or unauthenticated users can vote.	Ensures the voting system is responsive and secure.
NFR-002	Performance	The system shall maintain acceptable response time when multiple Club Members access discussions and polls simultaneously.	Medium	Pass: Benchmark testing under simulated peak load remains within the target response time. Fail: Response time exceeds the target under peak load.	Ensures the portal remains usable when many members participate simultaneously.

UML Use Case Diagram:


Book Club Reading Challenge & Discussion Portal
Log Daily Reading Progress
Club Member
Participate in Chapter Discussion
<<include>>
Mask Spoiler Content
Cast Monthly Poll Vote
<<include>>
Verify User Session
Flag Inappropriate Post
<<extend>>
Moderate Discussion Thread
Discussion Lead
Create Monthly Poll

Use Case Flow Specification:

Use Case: Vote in Monthly Book Poll
Primary Actor: Club Member

Preconditions

1. The Club Member is registered/logged into the portal.
2. A monthly book poll is currently active.
3. The member has not already voted in the current poll.

Postconditions

1. The member's vote is recorded.
2. The member cannot vote again in the same poll.
3. The poll tally is updated.

Main Success Scenario

1. Club Member opens the active monthly book poll.
2. System displays the available book options.
3. Club Member selects one book.
4. Club Member submits the vote.
5. System verifies that the member has not already voted.
6. System records the vote.
7. System updates the poll tally.
8. System confirms that the vote was successfully submitted.

Alternate Flow: Duplicate Vote

1. Club Member selects a book and submits a vote.
2. System detects that the member has already voted in the current poll.
3. System rejects the new vote.
4. System displays a message informing the member that they have already voted.
5. The existing vote remains unchanged.